Graphic Era deemed to be University DEHRADUN

PROJECT AND TEAM INFORMATION

Project Title

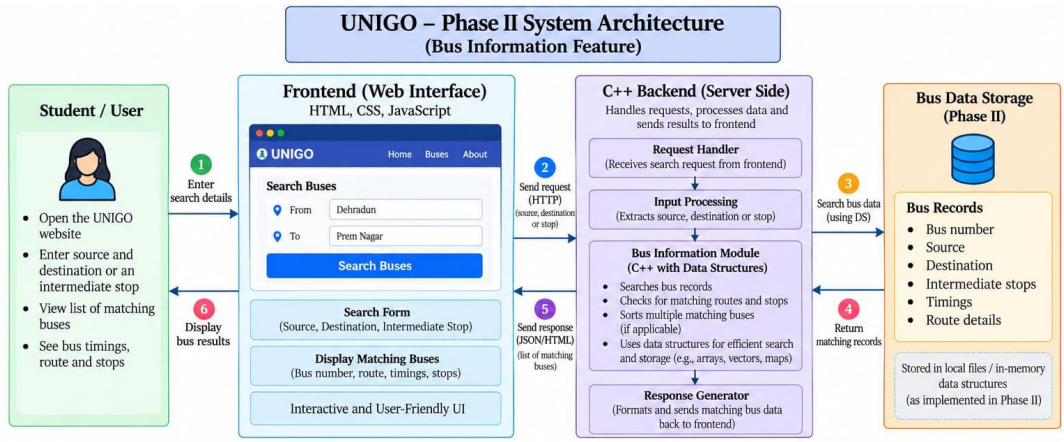
UNIGO-A Smart Campus Navigation and Student Assistance Platform

Team Name: *Team #*	*UNIGO*
Team member 1 (Team Lead)	*Yadav, Sapna – 2510011587* *sap.na0515ydv@gmail.com* 
Team member 2	*Vanshika – 2510012685* *vanshikaparmar0729@gmai* 
Team member 3	*Nahar,Srishti – 2510014554* *srishtinahar555@gmail.com* 
Team member 4	*Pun,Anushka – 2512510037* *anushkapun1505@gmail.com* 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

UNIGO is a website aimed at making it easier for students to find the right college bus. Students often have difficulties because they are not familiar with route numbers, the stops, or the shift times. I n Phase 2, the Bus Information feature was developed, so that students can input a destination or a bus number and then see all the buses that match their search, together with the route, the stops, the departure time, and the details regarding the campus gate. The frontend part of the system makes use of HTML, CSS, and JavaScript, while the backend is written in C++ and includes a bus-search module written in C. This module holds the bus records in structures and arrays, matches them to the students' searches, and arranges the results according to departure time. We have prepared a synthetic dataset containing the route numbers, shift times, departure times, destinations, and stops, and have tested it using sample inputs. The main goal is to give students quick and easy access to bus route information. In the future, the system will allow students to find their bus on campus when returning.



Updated Project Approach and Architecture

The system is made up of three layers: the frontend, which is composed of HTML, CSS and JavaScript, gathers the students' search queries and shows the results; the C++ backend handles the request, checks the input and passes it on to the bus module; and the C module is responsible for storing the bus records, carrying out searches and arranging the matches in order of departure time. The search facility looks at both the final destinations and the intermediate stops and also allows you to query by bus number. When several buses match the search criteria, the results are shown in a consistent order even if the departure times are the same. Before displaying the results, the program handles cases where the search is empty, the destination is unknown, the bus number is invalid, or the departure time is missing. The bus data is a synthetic dataset which has been manually checked for accuracy; the code can be found on GitHub. The C module and the C++ backend need to be compiled and linked. The following step is to connect the frontend to the backend and then test the entire search process from input through to result.

Tasks Completed (7 pts)

Task Completed	**Team Member**
Frontend development using HTML, CSS, and JavaScript	Sapna Yadav
Bus data preparation, organization, and testing support	Vanshika
C++ backend development and integration	Srishti Nahar
C module development using data structures, searching, and sorting	Anushka Pun

Challenges/Roadblocks

Several challenges were encountered during the development of the UNIGO website, as outlined below:
1. The connection between the frontend and the backend was aimed at making it possible for the website to send search requests to the C++ backend and correctly show the results that were returned.
2. The C module and the backend had to be compiled and linked properly so that the backend could use the bus-searching function which was written in C.
3. When looking up by destination or by bus number: although most students know their destination, some students only know the bus number. In order to cater for both situations, the search option has to check all the route stops and the intermediate stops, not just the final destination.
4. When several buses are serving the same destination: The system must show all the buses that match, indicating their numbers and destinations, and it should sort them in a consistent manner, even in instances where the buses have the same departure or arrival times.
5. How accurately the bus information is given: The bus numbers, the route stops, the departure times, and the details regarding the campus gates must all be carefully arranged in ascending order and manually checked.
6. When it comes to invalid inputs and missing data, the website should deal with cases such as empty searches or unknown destinations and with invalid bus numbers and records that have unavailable departure times before giving out any incorrect results.

Tasks Pending

Task Pending	Team Member
Develop the frontend pages and interfaces for the campus map and navigation, the Lost and Found service, and the student reviews and feedback system.	Sapna Yadav
Develop backend operations for new features, implement login and authentication, and connect the website to the database.	Srishti
Develop graph-based route-stop management for bus features, including insert and delete stop logic, and assist with the other three features.	Anushka
Organize campus location information, Lost and Found records, and review-related data; assist with validating and testing new features.	Vanshika

Project Outcome/Deliverables

Unigo is a website that has been proposed in order to help students find the right bus routes and obtain information about the campus. At the moment, our focus is on the bus route feature since a great many students have trouble in choosing the correct bus. Through the website, users can enter their destination and then see the relevant route numbers. The platform has been developed using HTML, CSS and JavaScript, with C++ being used for the backend processing. A synthetic dataset is used to provide the route numbers, the shift times and the details of the routes. Arrays and structures are used to store the bus routes, and search algorithms have been put into place to match the routes to the user's input. The main aim of our project is to improve the accessibility of bus route information for students.

Progress Overview

The students found it difficult to identify the right bus routes, to manage their shift times, and to choose suitable destinations. As a result, the main thing we have had in mind at this stage has been the development of the bus information feature. We have produced a synthetic dataset which contains the shift times, the route numbers, the departure times, the destinations, and the bus stops. We uploaded the project to GitHub, and we are working together to improve its operational efficiency. The frontend was written using HTML, CSS and JavaScript, and we are going to implement route-search functionality by means of C++ concepts. At the moment we are looking at the coding requirements and working on the fundamental system functions. The principal difficulty has been getting the backend to work with the frontend and setting up the route-search logic in the web system. We have carried out some initial tests using sample inputs in order to examine the system's capability of identifying the correct bus routes. We are currently looking at the results to find any errors and to improve accuracy.

Codebase Information

- The project has been uploaded to GitHub, including both frontend and backend code.
- Code commits have been made according to the progress achieved so far.
- A commit has been made for uploading the synthetic dataset.
- Separate commits have also been made for frontend and backend development.
- GitHub repository: `https://github.com/Anushkapun150507/PBL_DSCPP_SEM3/tree/main`
- Problem identification and planning — Completed.
- Synthetic bus route dataset — Prepared and uploaded.
- Frontend development — HTML, CSS and JavaScript.
- Backend development —C++.
- GitHub repository and commits — Completed.
- Route searching based on user input — Completed.
- Frontend and backend integration — Completed.
- Complete testing and validation — In progress.
- Initial testing has been performed using sample inputs to check the bus route-searching functionality.
- Basic testing of the frontend, backend, and synthetic dataset is in progress to check their working and integration.

Testing and Validation Status

Test Type	**Status (Pass/Fail)**	**Notes**
Manual functional testing(Bus Guide)	Pass	We searched different destinations and checked that the departure point and timings shown were correct.
User interface testing (website)	Pass	We opened the website and clicked through the pages and buttons to check that they work and respond properly.
Other features (map, lost and found, reviews)	Not tested yet	These features are still in progress, so we will test them once they are ready.

Deliverables Progress

- Campus map and navigation: in progress
- Bus guide: completed, working on the website
- Lost and found: in progress
- Reviews and ratings: in progress
- Database and website: in progress,

the website frontend is ready and data set for bus navigation is ready